

Minimal Structural Sufficiency and the Pruning Paradox in Neural Networks

Abstract

Modern neural networks exhibit extreme parameter redundancy: up to 90% of weights can be removed without immediate loss of performance. This empirical fact is well documented through the Lottery Ticket Hypothesis, structured sparsity, and hardware-aware pruning techniques. However, performance does not degrade smoothly with pruning. Instead, it remains stable across wide removal regimes and then collapses abruptly.

This work argues that sparsity alone cannot explain this behavior. We introduce the concept of **minimal structural sufficiency** ($F_{\min}$): *the minimal, non-distributable configuration required for stable inference. Pruning succeeds not because models are sparse, but because it preserves $F_{\min}$.* Failure occurs when pruning violates this structural threshold, triggering abrupt throughput collapse. This framework provides a unifying explanation for pruning dynamics, failure modes, and efficiency limits in minimal-resource machine learning.

1. Introduction — The Pruning Paradox

Several empirical facts about modern neural networks are broadly uncontested:

- Overparameterized networks contain massive redundancy.
- Large fractions of parameters can be removed with little or no loss in accuracy.
- Structured sparsity enables significant inference speedups.
- Performance remains stable across wide pruning ranges.

Yet an unresolved problem persists:

Why does performance remain stable until it suddenly collapses?

Existing pruning literature describes *how* parameters can be removed efficiently, but does not explain *what must remain* for inference to remain viable. Accuracy curves do not degrade proportionally to parameter count. Instead, they exhibit plateau-like behavior followed by sharp transitions.

This paper argues that pruning behavior is governed not by sparsity levels, but by the preservation or violation of a minimal structural core.

2. Limits of Sparsity-Based Explanations

Sparsity is often treated as a proxy for efficiency. However, sparsity alone is insufficient as an explanatory principle.

- Two networks with identical sparsity ratios can exhibit radically different performance.
- Hardware-aware pruning improves execution speed but does not explain inference stability.
- Magnitude-based criteria correlate weakly with functional importance.

Sparsity measures *how much* is removed, but says nothing about *what cannot be removed*.

The key variable is not parameter count, density, or compression ratio, but **structural sufficiency**.

3. Minimal Structural Sufficiency ($F_{\min}$)

We define **minimal structural sufficiency**, denoted ($F_{\min}$), as:

■ *The minimal structural configuration required for stable inference on a given task.*

Key properties of ($F_{\min}$):

- **Non-distributable** — it cannot be arbitrarily spread across parameters; certain structures are irreducible.
- **Non-recoverable once breached** — once violated, retraining cannot reliably restore performance.
- **Architecture-relative but task-invariant** — different architectures may realize different ($F_{\min}$), but the task imposes a fixed structural requirement.
- **Independent of parameter count** — large and small models may share the same effective ($F_{\min}$).

Pruning succeeds because it removes non-contributing substructures while preserving ($F_{\min}$).

4. Throughput Collapse and the Role of (Ψ)

To operationalize sufficiency, we use **operational throughput**, denoted (Ψ) , following prior work on measurable throughput in open systems.

- Above $(F_{\min})$: (Ψ) degrades gracefully.
- At $(F_{\min})$: the system operates at maximal efficiency.
- Below $(F_{\min})$: (Ψ) collapses abruptly.

This explains why pruning curves exhibit **phase-transition-like behavior**, analogous to percolation thresholds, rank collapse in linear systems, or loss of spectral support.

Retraining cannot reliably restore performance once $(F_{\min})$ is violated, because the structural support required for inference no longer exists.

5. Pruning as a Sacrificial Necessity

Pruning is often framed as optimization or compression. We propose a different interpretation:

Pruning is a structural survival mechanism.

In effective systems, non-contributing substructures are actively shed to preserve coherence. In machine learning, this manifests as magnitude pruning, block sparsity, and neuron death during training.

Efficient models survive not because they are sparse, but because they have already converged to a minimal sufficient structure.

6. Relation to the Lottery Ticket Hypothesis (Without Mythology)

The Lottery Ticket Hypothesis (LTH) demonstrated that small subnetworks can match the performance of full models when initialized correctly.

We reinterpret LTH as follows:

- LTH **describes discovery**.
- $(F_{\min})$ **explains necessity**.

Winning tickets are not lucky subnetworks; they are instances of minimal sufficiency uncovered through overparameterization.

7. Failure Modes

Observed failure modes align naturally with violations of $(F_{\min})$:

- **Over-pruning** → loss of global modes.
- **Structured pruning in the wrong basis** → breaks minimal structure.
- **Late-stage pruning** → higher risk due to rigidified representations.

These failures are structural rather than stochastic.

8. Implications for Minimal-Resource Machine Learning

This framework yields direct implications for efficient ML:

- Training with **10–20% capacity** is feasible if structure is preserved.
- **Continuous learning** is more robust than static training followed by pruning.
- Hardware-aware design should target preservation of $(F_{\min})$, not maximal sparsity.
- Efficiency gains plateau once pruning approaches structural limits.

Minimal-resource ML is not about smaller models, but about **structurally sufficient ones**.

9. Conclusion

Neural networks do not fail because they lose parameters.
They fail when they lose structure.

Pruning succeeds when it removes expendable mass, and fails when it violates minimal sufficiency. Understanding this boundary is essential for the next generation of efficient machine learning systems.

Throughput vs Pruning Ratio Diagram

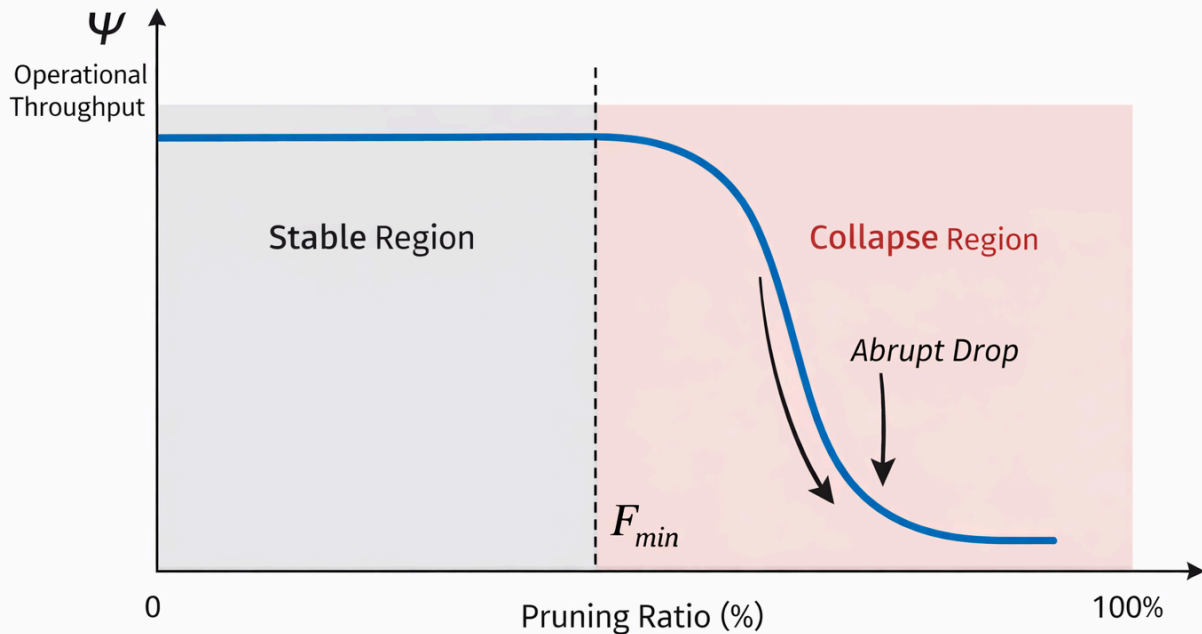


Figure 1: Throughput vs. Pruning Ratio

Figure 1 — Phase Diagram of Throughput Under Pruning

Operational throughput (Ψ) as a function of pruning ratio. The stable regime corresponds to pruning that removes redundant structure while preserving (F_{min}). The vertical boundary marks the (F_{min}) threshold. Beyond this point, throughput collapses abruptly and irreversibly, indicating structural failure rather than quantitative degradation.

Appendix — Mapping to ISSP and Spectral Interpretation

This appendix connects **minimal structural sufficiency** (F_{min}) with the **Induced Structural Stability Principle (ISSP)** and provides a spectral interpretation.

ISSP states that under constraints, functional behavior concentrates into a reduced set of structurally stable modes. In neural networks, architectural, data-induced, and hardware constraints drive this concentration.

Within this framework, $(F_{\min})$ corresponds to the minimal subset of dominant modes required to sustain non-zero operational throughput ($\Psi > 0$).

From a spectral perspective, pruning removes weak or redundant modes with little effect until a dominant eigenmode is destabilized. At that point, effective rank collapses and ($\Psi \rightarrow 0$).

ISSP explains *why* structure concentrates, $(F_{\min})$ defines *what must remain*, and (Ψ) measures *whether the system still functions*. Together, they account for the abrupt and irreversible failure observed under excessive pruning.